

VALEN LI

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EDUCATION

Julia R. Masterman High School

GPA: 4.53/4.0

Robotics Team (Co-Founder & Lead Programmer) | Chess Team (Co-Captain) | AI Club (President)

Philadelphia, PA

Sep 2018 – June 2026

The Wharton School of the University of Pennsylvania

GPA: UW- 4.0/4.0

STAT 0001: Intro to Statistics and Data Science

Remote

Sep 2024 – Mar 2025

Community College Of Philadelphia

Courses: MATH 163: Discrete Mathematics I — Math 263 Discrete Mathematics II

In-Person

Sep 2025 – Jun 2025

The Wharton School of the University of Pennsylvania

3-week residential program at UPenn called: The Leadership in the Business World

Residential

Jul 2025 – Aug 2025

Oberlin College

Pioneer Research Program (Under Full-Scholarship and given Four College Credits)

Remote

Feb 2025 – September 2025

EXPERIENCE

Julia R. Masterman Secondary School

Technology Teacher Leader Aide / IT

Philadelphia, PA

Sep 2025 – Present

- Managed Chromebook deployment and configuration for students
- Provided technical support assistance to teachers and students, resolving hardware and software issues
- Assisted teachers with laptop repairs, inventory indexing, cable management, and grading papers

Wharton's Stevens Center for Innovation in Finance

Full-Stack App Developer

Philadelphia, PA

Jun 2024 – Jun 2025

- Enhanced key features of a student loan application to accurately project college costs, using PostgreSQL for data management and Spring Boot for API development
- Designed a responsive website for the Finiverse project, and created web features for a data visualization platform.

Root2Success

Co-President

Philadelphia, PA

Feb 2022 - Present

- Tutored over 50 hours and developed comprehensive curricula to teach mathematics to over 250 students across more than 25 partner schools in the School District of Philadelphia.
- Led a team of over 80 high school volunteers and organized in-person competitions to promote engagement and foster a love for mathematics.
- Redesigned and developed the [website](#) using React and Node.js

Carnegie Mellon University

Cyber-Security Researcher

Remote

Feb 2025 – August 2025

- Worked with professor Hanan Hibshi in the research concentration of: The Practical Challenge of Writing Secure Code
- Contributed to a research project on cybersecurity, focusing on using Self-Adapting Language Models (SEAL) for improving email phishing filters.
- Developed a research proposal and methodology for creating an AI-powered filter that can autonomously learn from new phishing tactics.

University of Washington

Machine Learning Researcher

Remote

June 2024 – Aug 2024

- Trained ML models such as XGBoost, LightGBM, CatBoost, Random Forest, K-Means Clustering, and GNNs on an emerging alternative (SMBJ) to traditional PCR techniques used in genetic sequencing and identification.
- Built an AI **system** to classify COVID-19 variants from Single Molecule Break Junction (SMBJ) conductance signatures; achieved up to 99.97% accuracy (CatBoost).
- Developed a preprocessing pipeline: filtering, R^2 -based QC, conductance conversion, and 1D/2D histogram features.

EXTRACURRICULAR ACTIVITIES

FIRST Robotics

Lead Programmer | Mentor | Inspection Lead | Tournament Volunteer | FRC Co-Founder

Philadelphia, PA
Sep 2022 – Present

- Lead Programmer of teams: 22284 and 22279 (state-qualifier) simultaneously
- Co-founded FRC team 10918 Bluesteel Dragons first year at Masterman since 2019
- Raised \$20k in funding to support new FRC team
- Created an autonomous program utilizing Virtual Field Mapping, Computer Vision, and Pure Pursuit algorithms
- Volunteered at FIRST Lego League (FLL) tournaments and mentor my school's FLL robotics team weekly
- Winner of: RoboJawn 2024 | Motivate Award | Think Award

USChess

Masterman Chess Team Co-Captain | Tournament Director | Chess Camp Instructor

Onsite
Sep 2018 – Present

- Co-Captain for the Elementary/Middle/High School Masterman Chess team
- Held weekly coaching sessions for the Elementary and Middle School
- Certified tournament director | Volunteered to direct the Elementary/Middle School Chess League
- Earned: >\$3,500 | 2x @ World Open (4th) | 3x @ Nationals (5th) | Liberty Bell (1st) | Chess Congress (4th)

PROJECTS

Company Case Competition Analysis | *Figma, Business Strategy*

Jul 2025 – Aug 2025

- Developed a comprehensive business strategy and pitch for Merck, recommending a partnership to create a complementary product for the newly acquired drug, Ohtuvayre.
- Designed and built a prototype for a smart inhaler and tracking app using Figma, visualizing the proposed product and user experience.
- Presented the pitch in a 15-minute presentation for the Leadership in the Business World case competition.

Real-time Edge Derivative Instance Segmentation | *Python, OpenCV, PyTorch, YOLOv11* May 2025

- Authored a mathematical paper on image derivatives (Sobel, Laplacian) and implemented OpenCV edge detectors with code examples.
- Built a real-time object detection and instance segmentation pipeline using YOLOv8 with FPS overlay and COCO class labels.
- Documented setup and usage; provided a reproducible environment and compiled LaTeX paper.

The Woodlands Cemetery Project | *Web Development, Research*

Mar 2025 – May 2025

- Researched and unearthed the life of James Hepburn Campbell, a 19th-century individual, as part of a rigorous research project for the A.P. United States History course.
- Independently created a professional website in just 3 days to detail Campbell's life and document the team's research process.
- Contributed over 280 hours to the project, demonstrating strong commitment and research skills.

AWARDS

Good Citizen Award

Mar 2025

Awarded by The Union League of Philadelphia for being 2/5,000 nominated by a Philly nonprofit for demonstrating hard work, honesty, curiosity, and loyalty

1st Place - Innovation League 2024

Dec 2024

Awarded \$5000 | Prototyped, developed, and pitched an innovative tech startup to a panel of judges.

2nd Place - Ctrl+Shift 2024

May 2024

Awarded \$2500 | Developed an innovative website that redefines AI education.

1st Place- AMC 12B with Distinction and AIME Qualifier

Nov 2024

Scored 1st in my school among seniors and juniors 2 years in a row and scored in the top 5% of all test takers in the country qualifying me for the American Invitational Mathematics Examination

3rd Place- Penn Mathematics Contest

April 2025

Awarded by The University of Pennsylvania for winning 3rd Place, scoring a 65/80 at the Penn Math Contest (PMC IV)

Congressional App Challenge Finalist

Nov 2024

Recognized by the House of Representatives and Madeleine Dean

Questbridge National College Match Finalist and Questbridge Prep Scholar

September 2025

Recognized by Questbridge for academic excellence, personal qualities, and financial need, and for being a competitive candidate for admission to their college partners.